जामिया मिल्लिया इस्लामिया جامعہ ملیّہ اسلامیہ
Jamia Millia Islamia
A Central University
(NAAC Accredited A++ Grade)

HiDMamba

A Hierarchical Distillation Cascade for Interpretable, Edge-Deployable Mamba-Based ECG Classification

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M.Tech CSE (Data Science) - Minor Project (MDS-393)

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Professor
Department of Computer Engineering
Jamia Millia Islamia

HiDMamba - hierarchical distillation for edge-deployable ECG classification.

The Clinical Problem



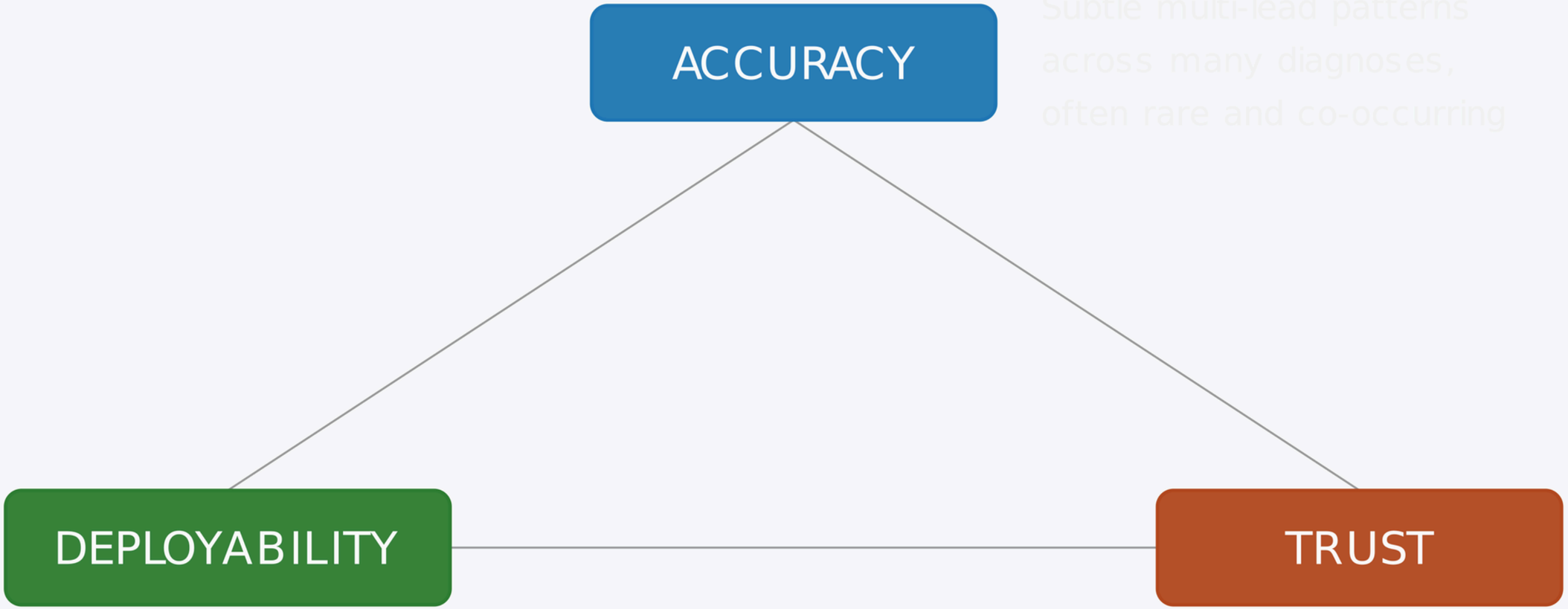
12-lead ECG

- ECG is the standard non-invasive tool for diagnosing cardiac conditions
- Automated ECG diagnosis is entering clinical and wearable use faster than trust can be established
- The models driving this shift are largely black boxes
- Cardiologists need support that is accurate, interpretable, and light enough for point-of-care use

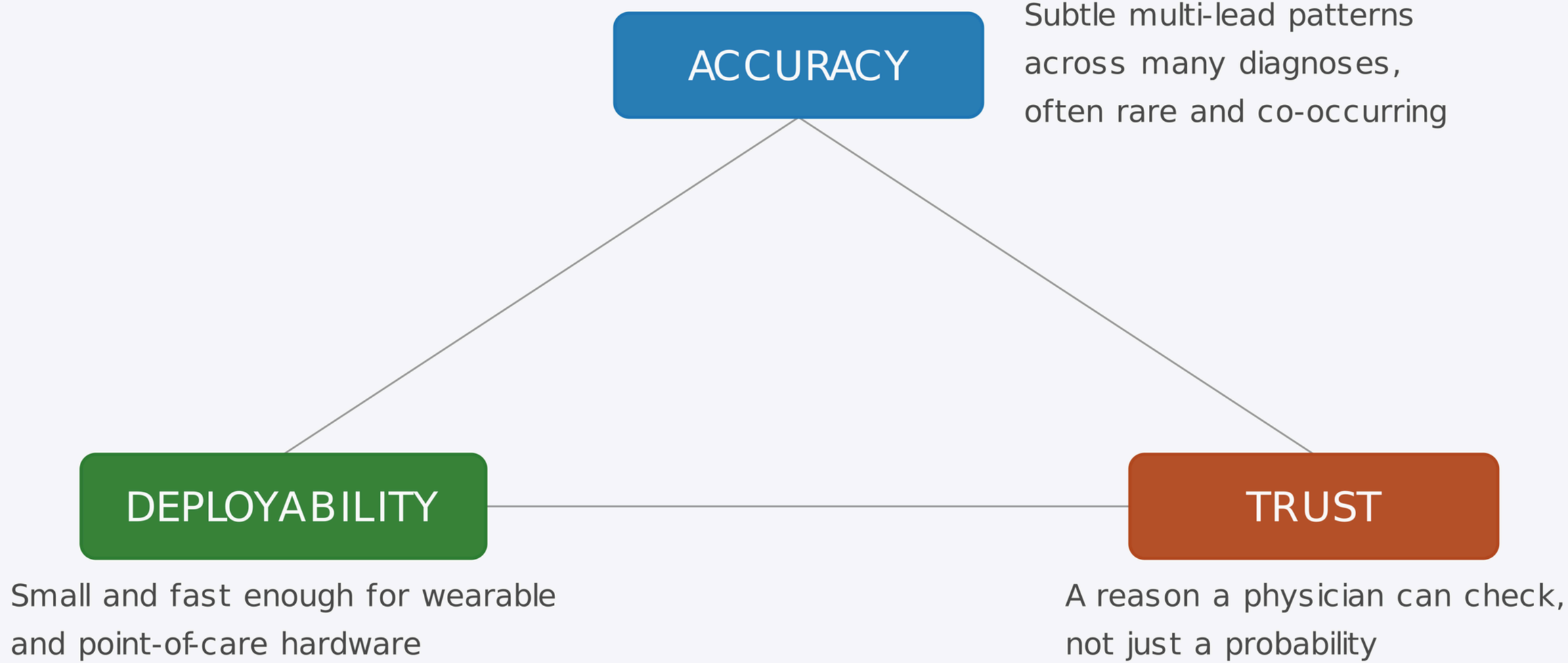
Automated ECG interpretation is entering practice ahead of trust.

ECG is central to diagnosis, but manual interpretation is a bottleneck.

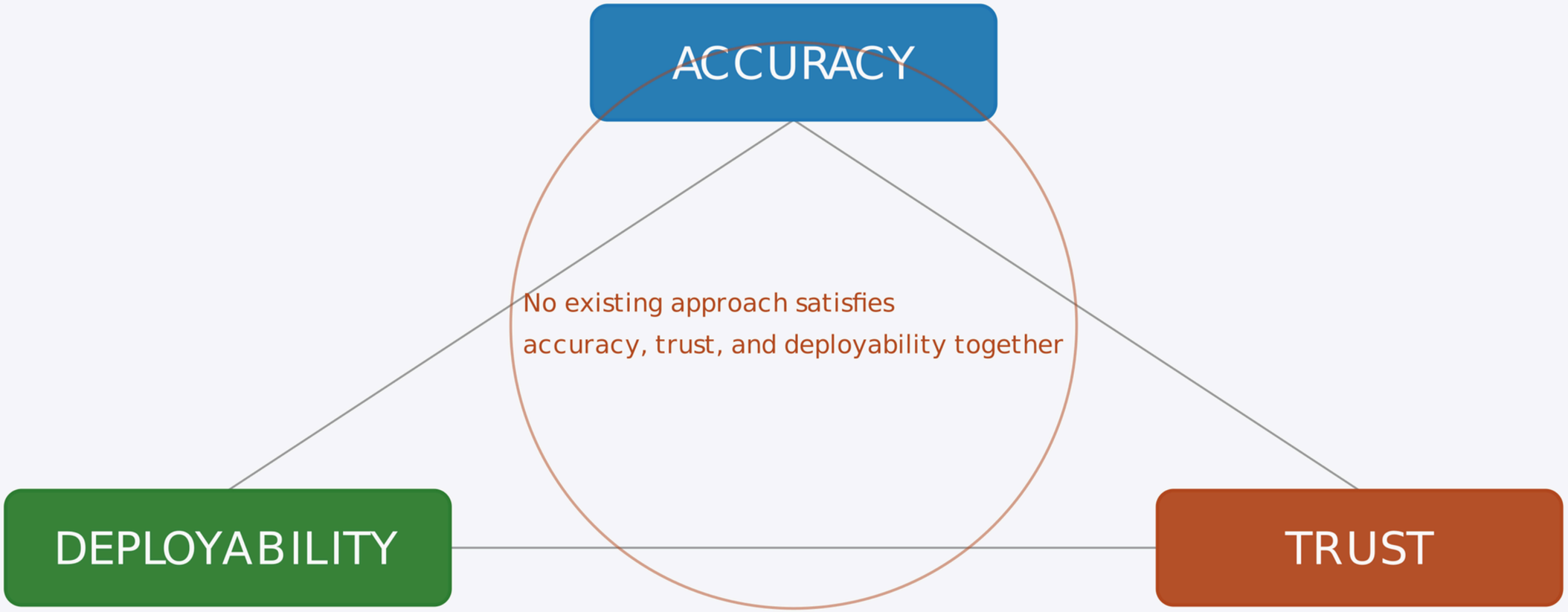
Accuracy, Trust, Deployability



Accuracy, Trust, Deployability



Accuracy, Trust, Deployability



The central challenge: ECG interpretation pulls a model in three directions at once.

Literature Review

ECG Datasets and Benchmarks

PTB-XL Dataset (Wagner et al., 2020)

- 21,837 clinical twelve-lead ECG recordings
- 18,884 patients
- 71 diagnostic statements (SCP-ECG standard)
- 5 superclasses, 23 subclasses
- Patient-stratified folds: 1-8 train, 9 val, 10 test

Literature Review

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Strodthoff et al. (2021) - Benchmarking Study

- First systematic cross-architecture benchmark on PTB-XL
- Best-performing architectures: ResNet-style and Inception-style
- Reports macro-AUC only, no macro-F1 figure exists

PTB-XL is the largest public 12-lead ECG dataset, the foundation of this work.

Literature Review

State Space Models: Mamba

Mamba: Linear-Time Sequence Modeling (Gu and Dao, 2023)

- Selective state space model (SSM)
- Input-dependent parameters: delta, B, C
- Near-linear-time sequence modeling: O(L) vs. O(L^2) for attention
- Hardware-aware scan: up to 40x faster than a standard implementation
- 5x higher inference throughput than Transformers

Matches Transformer-level quality with linear-time complexity

Mamba gives near-linear-time sequence modeling without attention's quadratic cost.

Literature Review

Mamba for Medical Time Series

ECGMamba (Qiang et al., 2024)

- Bidirectional SSM (BiSSM) applied directly to ECG classification
- Aims at efficient inference for the classification task
- No distillation component

MedMamba for Time Series (He et al., 2026)

- Principle-driven, multi-scale bidirectional Mamba
- Inductive biases: spatial centralization, multi-timescale composition, non-causal dependency
- Built for ECG and EEG time series specifically

Distinct from Yue and Li's MedMamba, a vision-Mamba model for 2D medical images

Existing Mamba-based ECG methods do not use knowledge distillation

Mamba has been adapted for ECG, but existing methods do not distill knowledge.

Literature Review

Self-Supervised Learning for ECG

3KG (Gopal et al., 2021)

- Physiologically-inspired augmentations
- VCG rotations and time masking
- Contrastive learning across leads
- 9.1% higher mean AUC than best SSL baseline at 1% of labeled data

CoRe-ECG (Qin et al., 2026)

- Unified contrastive + reconstructive pretraining
- Frequency Dynamic Augmentation (FDA)
- Spatio-Temporal Dual Masking (STDM)
- Combining both objectives outperforms either alone

Mehari and Strodthoff (2022)

- Benchmarks SimCLR, BYOL, CPC-style pretraining on ECG
- Best method: linear-eval performance only 0.5% below fully supervised
- Establishes the feasibility of SSL for ECG

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DEAN Pretraining

- Combines masked reconstruction with lead-wise contrastive learning

Self-supervised pretraining (contrastive + reconstructive) builds rich ECG representations.

Literature Review

Knowledge Distillation: Single Teacher to Cascade

Hinton et al. (2015)

- Soft targets via temperature scaling
- Transfers dark knowledge beyond hard labels

Relational KD (Park et al., 2019)

- Matches Gram-matrix relational structure, not raw activations
- Invariant to basis changes, critical for Mamba's state space

Multi-Label KD (Yang et al., 2023)

- Per-label sigmoid distillation, not softmax
- Naive softmax distillation fits multi-label targets poorly

Teacher Assistant KD (Mirzadeh et al., 2020)

- Student performance drops when the teacher-student gap is too large
- Chaining intermediate teachers closes the gap

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Single teacher -> Relational KD -> Multi-label -> Teacher Assistant -> Our cascade

Distillation compresses large teachers into compact students, but the capacity gap matters.

Literature Review

Interpretability and Prototype Learning

ProtoECGNet (Sethi et al., 2025)

- Prototype-based learning for ECG classification
- Multi-label, full 71-statement diagnostic task
- Case-based explanations from real ECG segments

HNSW / FAISS (Malkov and Yashunin, 2020)

- Efficient approximate nearest-neighbor search
- Hierarchical Navigable Small World graphs
- FAISS IndexFlatIP for CPU-resident retrieval

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Interpretability via Prototype Retrieval

- No post-hoc saliency maps needed, explanations are faithful by design

Prototype-based models give case-based explanations grounded in real ECGs.

Literature Review

Hierarchical Label Modeling

Wang and Hauskrecht (2023)

- Conditional tree-structured Bayesian network
- Models PTB-XL's 5-superclass, 23-subclass dependencies explicitly
- Improves multi-label classification via hierarchy constraints
- Exact inference, linear complexity in the number of classes

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↓

Our Hierarchical Distillation Cascade (HODs + Professors)

Explicitly modeling the label hierarchy improves consistency and interpretability.

The Research Gap

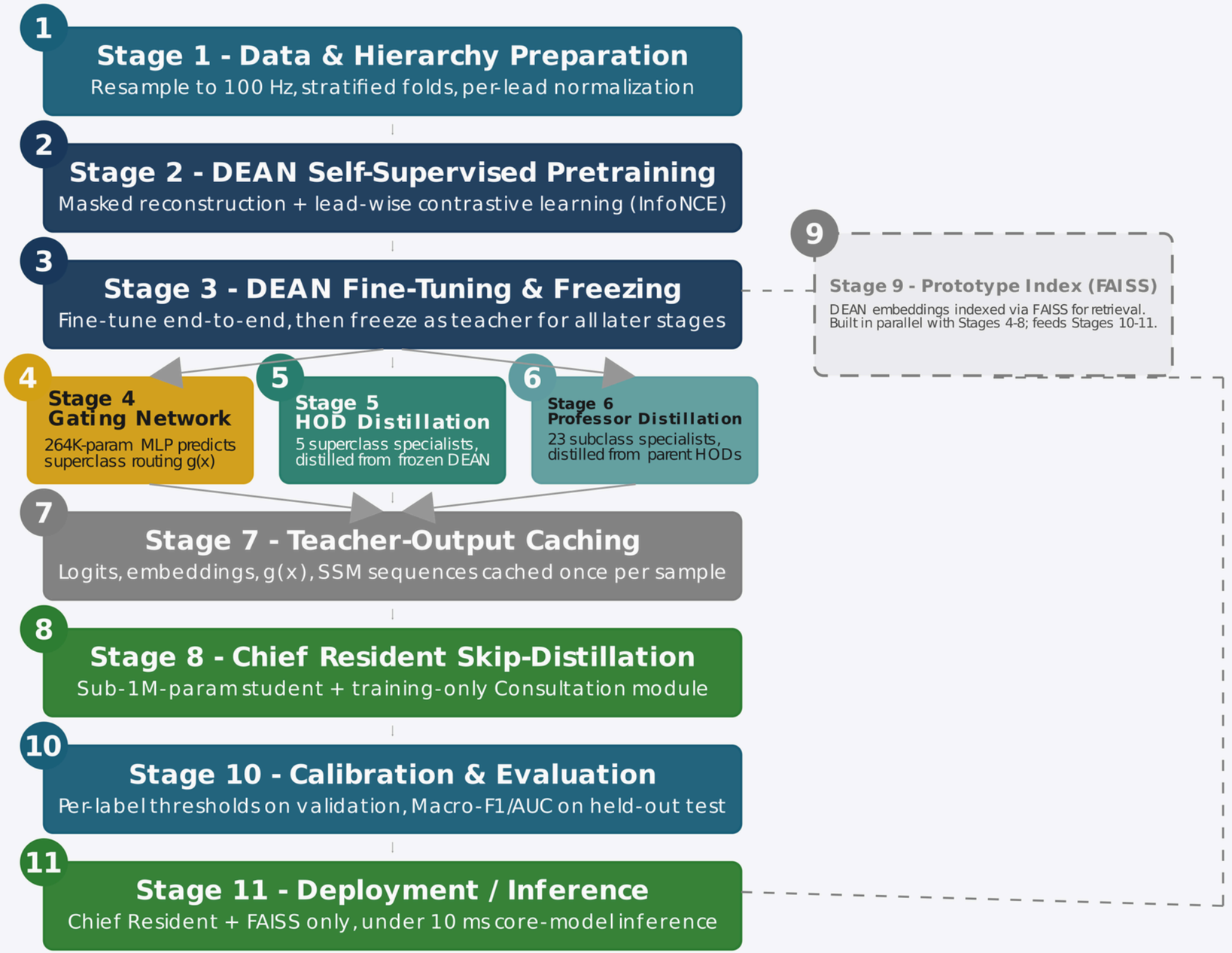
Does distilling along the actual clinical hierarchy - superclass first, then subclass - give a better-justified, more interpretable path to a deployable model than grouping teachers by capacity?

- Prior ECG distillation groups teachers by model size or capacity (HMT-KD, 2026)
- Prior ECG distillation ignores the diagnostic ontology cardiologists already use
- Hypothesis: hierarchy-aware distillation improves both interpretability and performance

The gap: existing ECG distillation groups teachers by capacity, not clinical ontology.

HiDMamba: Proposed Methodology

End-to-End Data Flow - Stage 1 (Preprocessing) to Stage 11 (Deployment)



HiDMamba - Hierarchical Distillation Cascade for Interpretable, Edge-Deployable ECG Classification

Methodology

HiDMamba: Four-Tier Cascade

DEAN (Mamba SSM Encoder, ~15M)

Tier 1 - Foundation. Learns electrophysiology from 21,837 recordings

↓

DEAN -> HOD -> Professor -> Chief Resident.

Builds on [1] [3] [11] [12] [13] [15] [17] [19] [21]

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HOD NORM | **HOD MI** | **HOD CD** | **HOD HYP** | **HOD STTC**

Tier 2 - Five superclass specialists. Which family owns this case

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NORM | AMI | IMI | LMI | PMI | LAnFB | LPFB | LBBB | RBBB | CLBBB | CRBBB | ILBBB

IRBBB | IVCD | LPR | LNGQT | LVH | RVH | 1STTC | 2STTC | 3STTC | 4STTC | 5STTC

Tier 3 - 23 subclass specialists. The specific diagnosis

DEAN -> HOD -> Professor -> Chief Resident.

Builds on [1] [3] [11] [12] [13] [15] [17] [19] [21]

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Tier 3 - 23 subclass specialists. The specific diagnosis

↓

Chief Resident (<900K)

Tier 4 - Bedside decision. Runs alone at inference

DEAN -> HOD -> Professor -> Chief Resident.

Builds on [1] [3] [11] [12] [13] [15] [17] [19] [21]

Stage 1

Data and Hierarchy Preparation



Resample to 100 Hz

Stage 1

Data and Hierarchy Preparation



→ Resample to 100 Hz

Stage 1

Data and Hierarchy Preparation



→ Resample to 100 Hz



Stage 1

Data and Hierarchy Preparation



→ Resample to 100 Hz

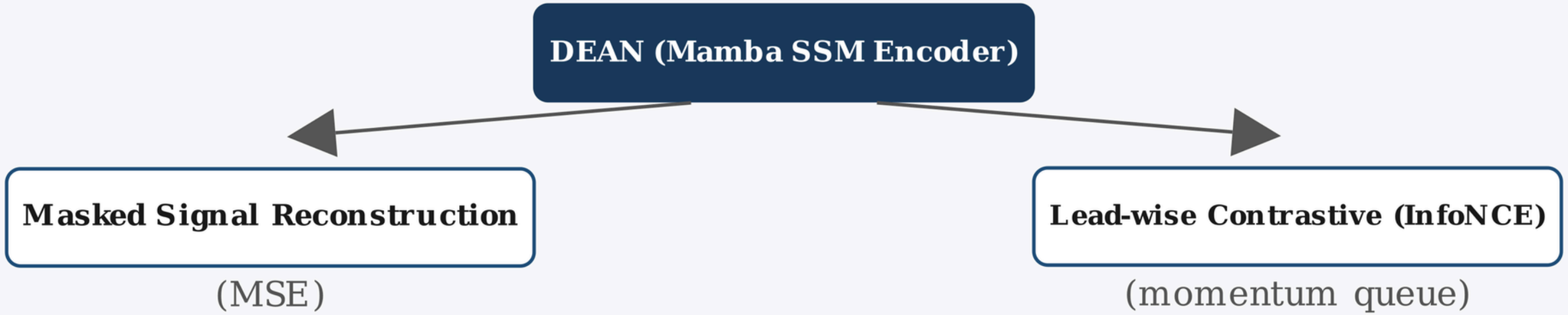


Folds 1-8: Train Fold 9: Validation Fold 10: Test

Compute per-lead normalization statistics

Stage 1: data preparation via resampling, stratification, and normalization.

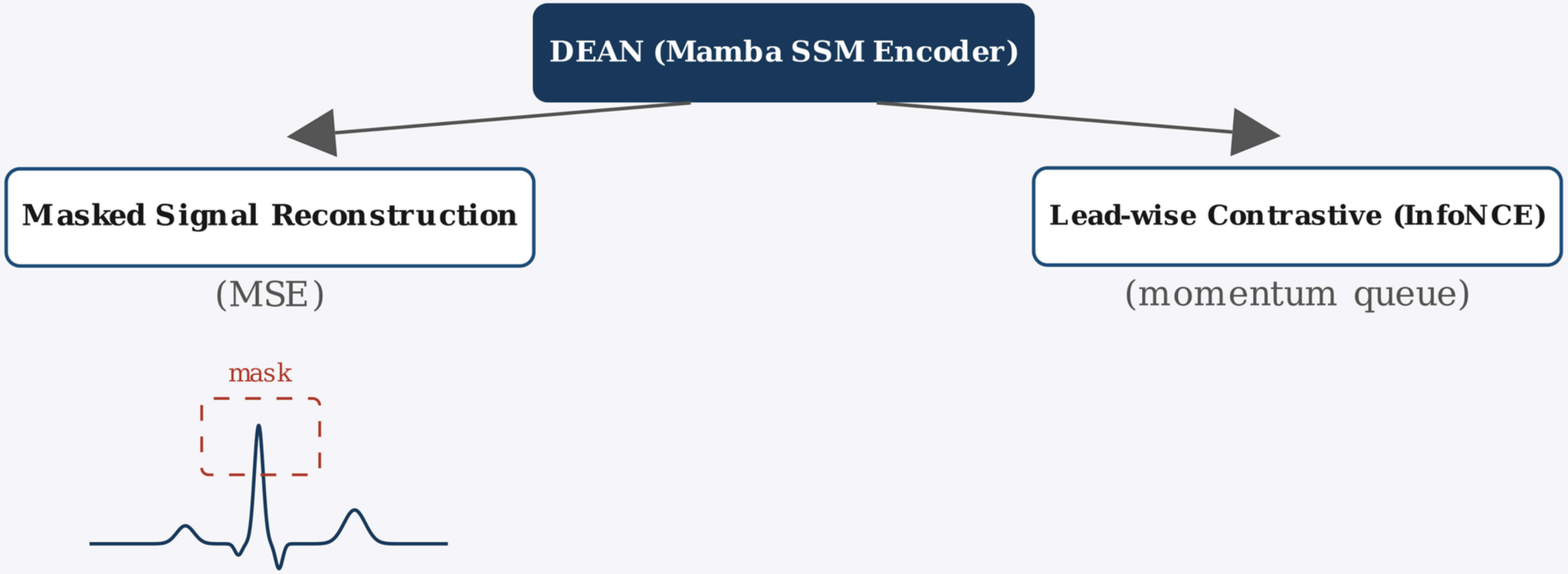
Stage 2: DEAN Self-Supervised Pretraining



Masked reconstruction + lead-wise contrastive learning.

[3] Gu & Dao, 2023 · [21] Kendall et al., 2018 · [22] He et al., 2020 · [23] van den Oord et al., 2018

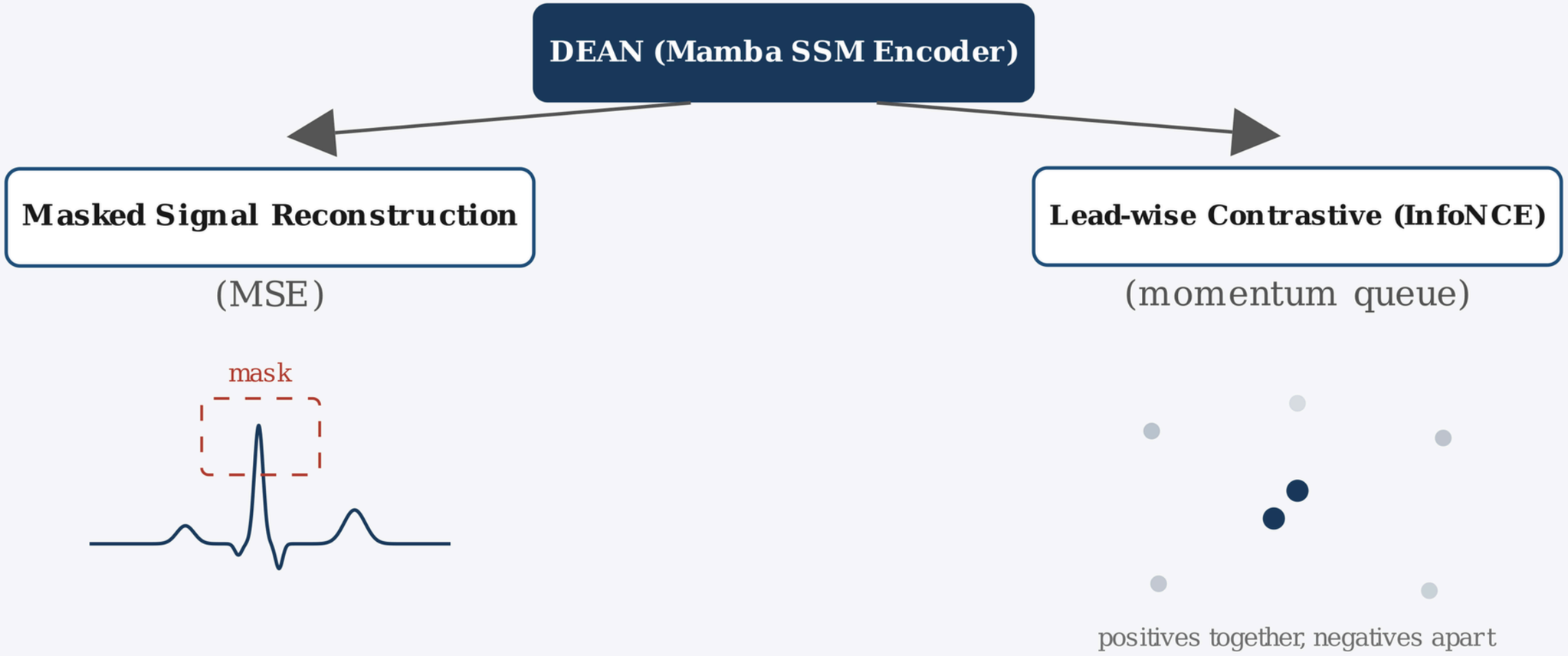
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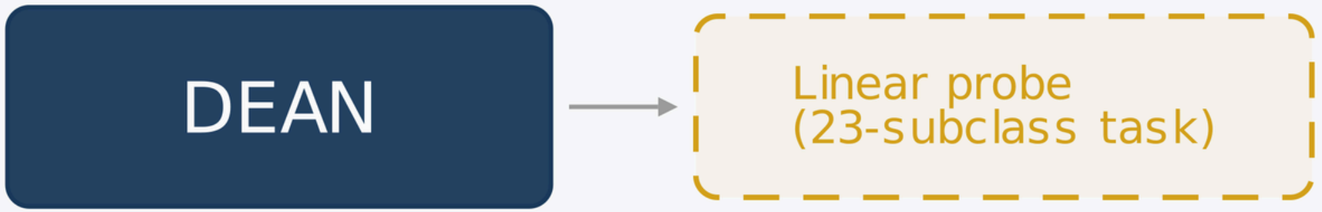
Stage 3

DEAN Fine-Tuning and Freezing

DEAN

Stage 3

DEAN Fine-Tuning and Freezing



Checks representation quality before fine-tuning

Stage 3

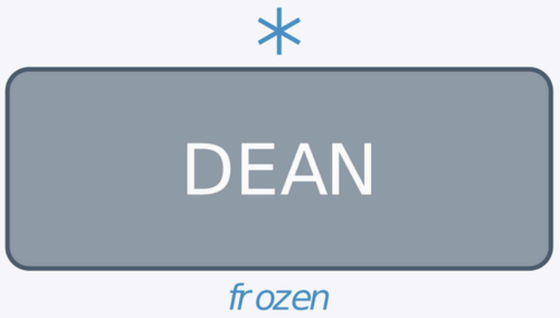
DEAN Fine-Tuning and Freezing

DEAN

Unfreeze and fine-tune end-to-end

Stage 3

DEAN Fine-Tuning and Freezing



Frozen - teacher only for every following stage

Stage 3: DEAN is fine-tuned, then frozen as the teacher for every later stage.

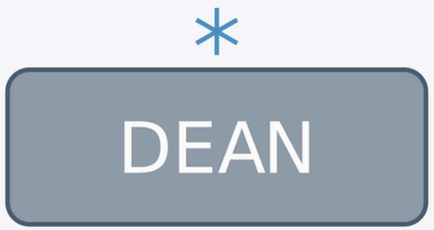
Stage 4

Gating Network

DEAN

Stage 4

Gating Network



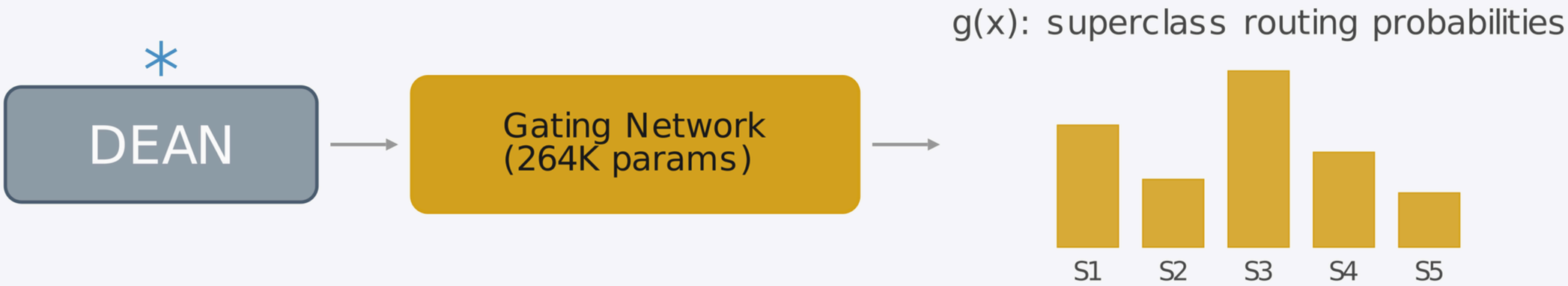
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Gating Network



Stage 4

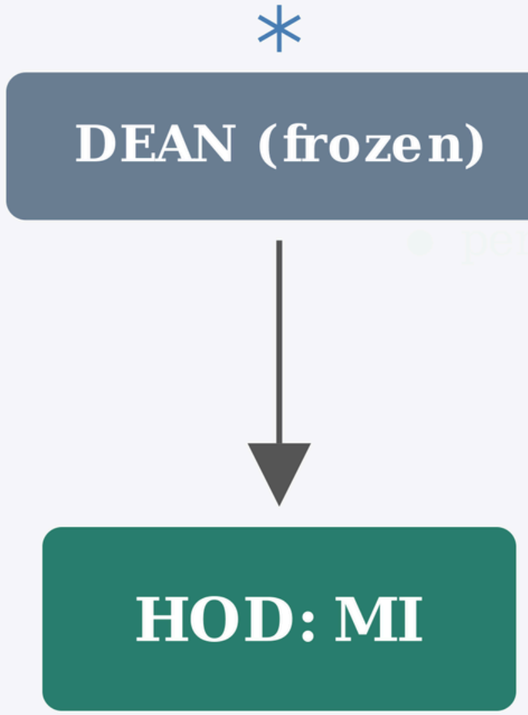
Gating Network



+ entropy regularizer, discourages collapse

Stage 4: the Gating Network predicts superclass routing probabilities.

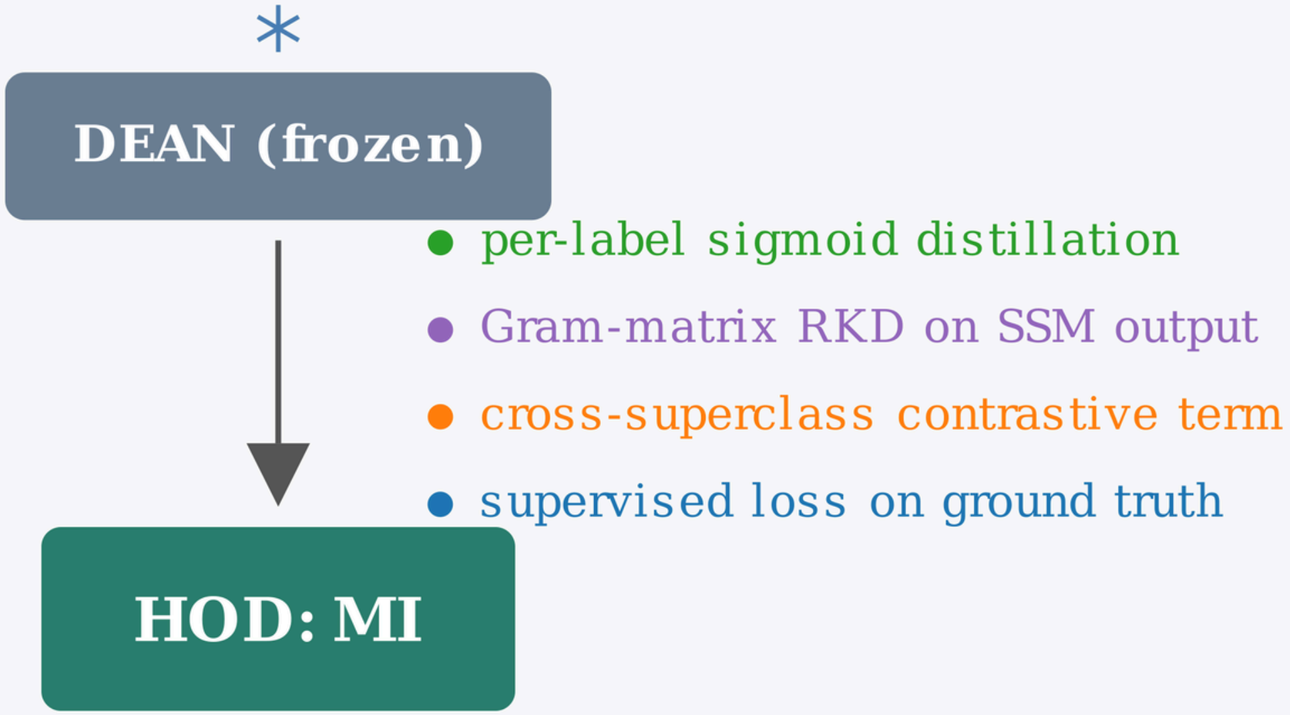
Stage 5: HOD Distillation



Five HODs, one per superclass, distilled from frozen DEAN.

[12] Park et al., 2019 (RKD) · [13] Yang et al., 2023 (multi-label KD)

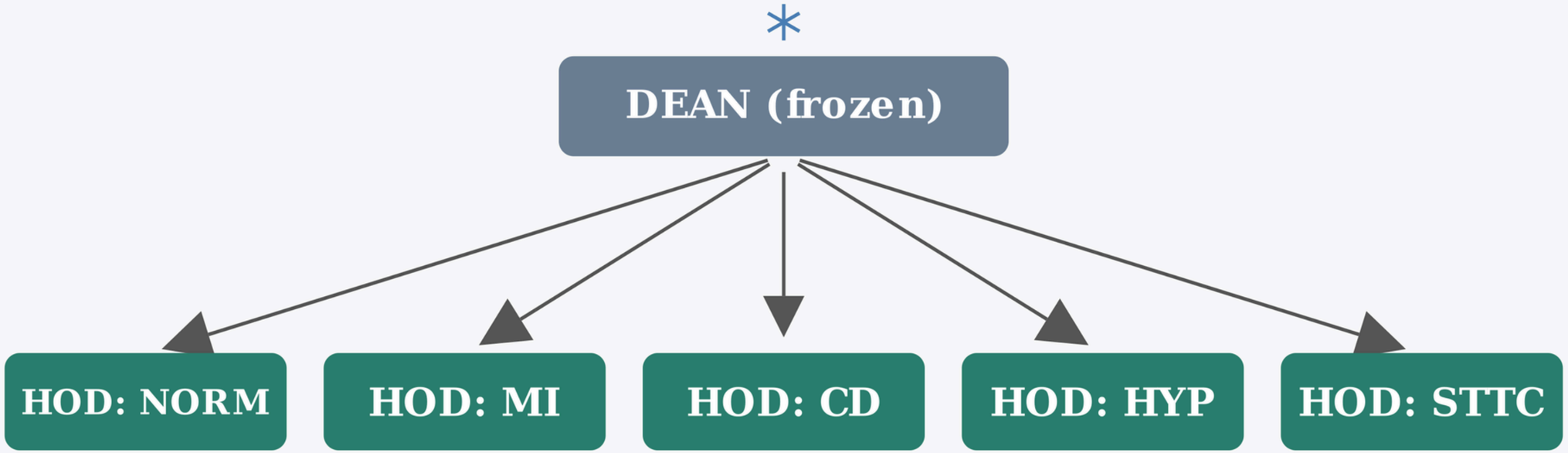
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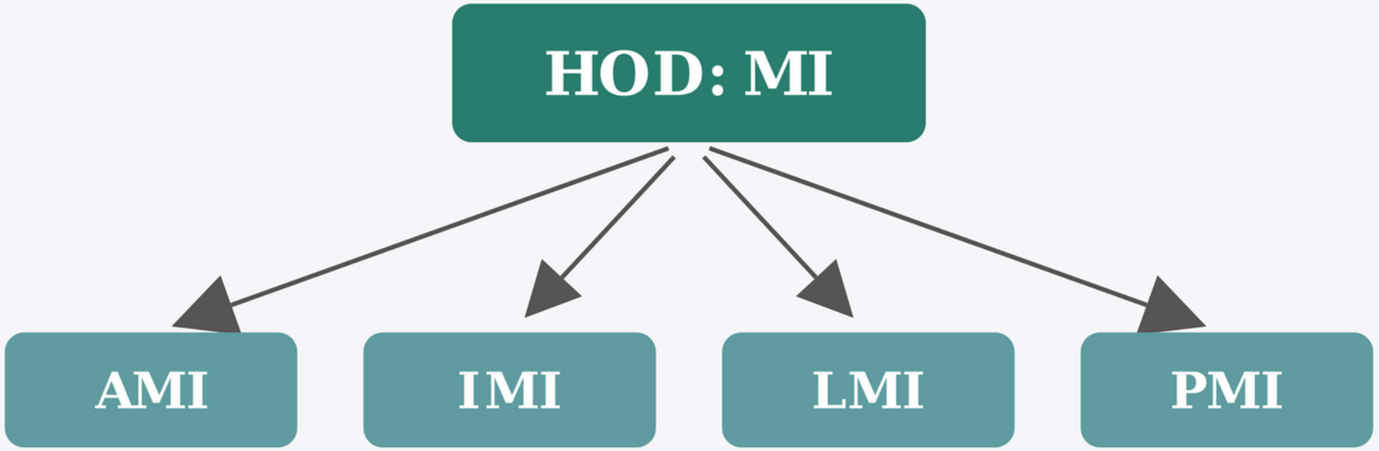


Same four-term recipe, replicated across five superclass specialists.

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Stage 6: Professor Distillation

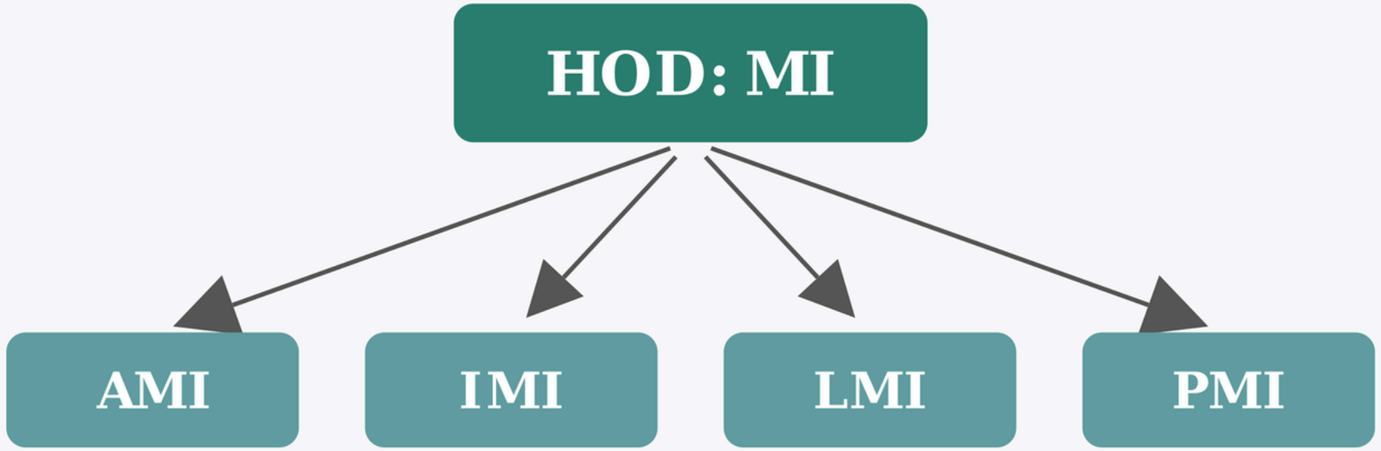


Hierarchy consistency: subclass probability <= parent superclass probability

23 Professors, one per subclass, distilled from parent HODs.

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Stage 6: Professor Distillation



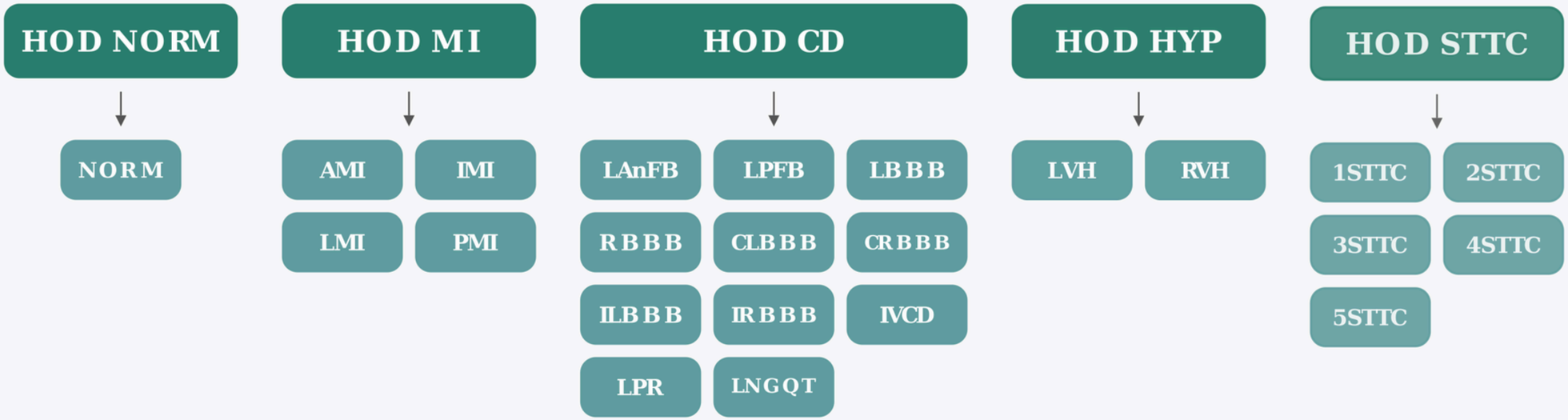
Hierarchy consistency: subclass probability <= parent superclass probability

Same distillation recipe as Stage 5, now sourced from the parent HOD.

23 Professors, one per subclass, distilled from parent HODs.

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Stage 7

Teacher-Output Caching

DEAN

Gating

5 HODs

23 Professors

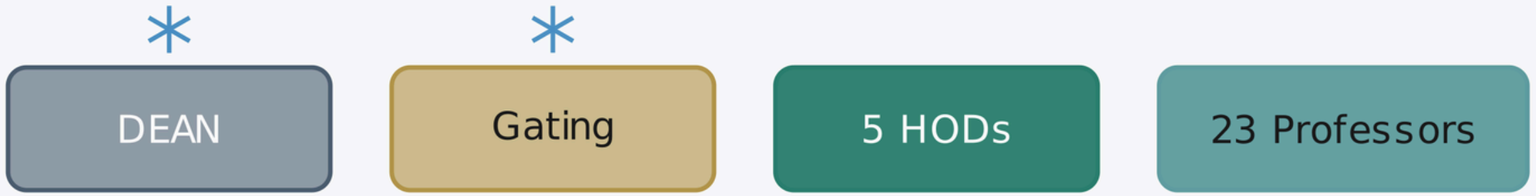
Stage 7

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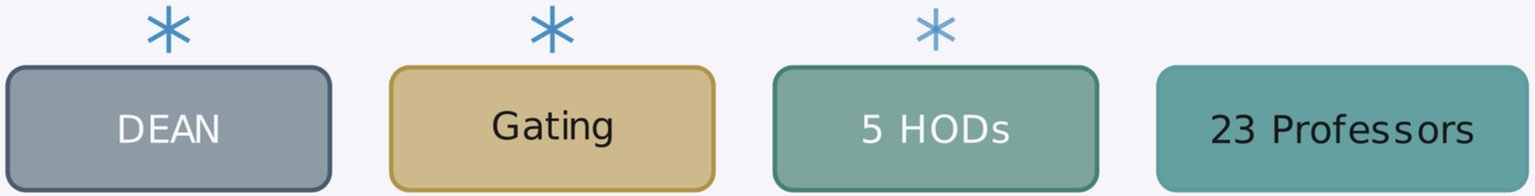
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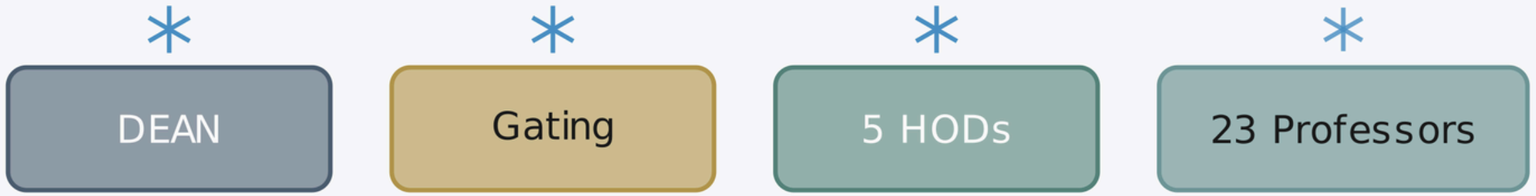
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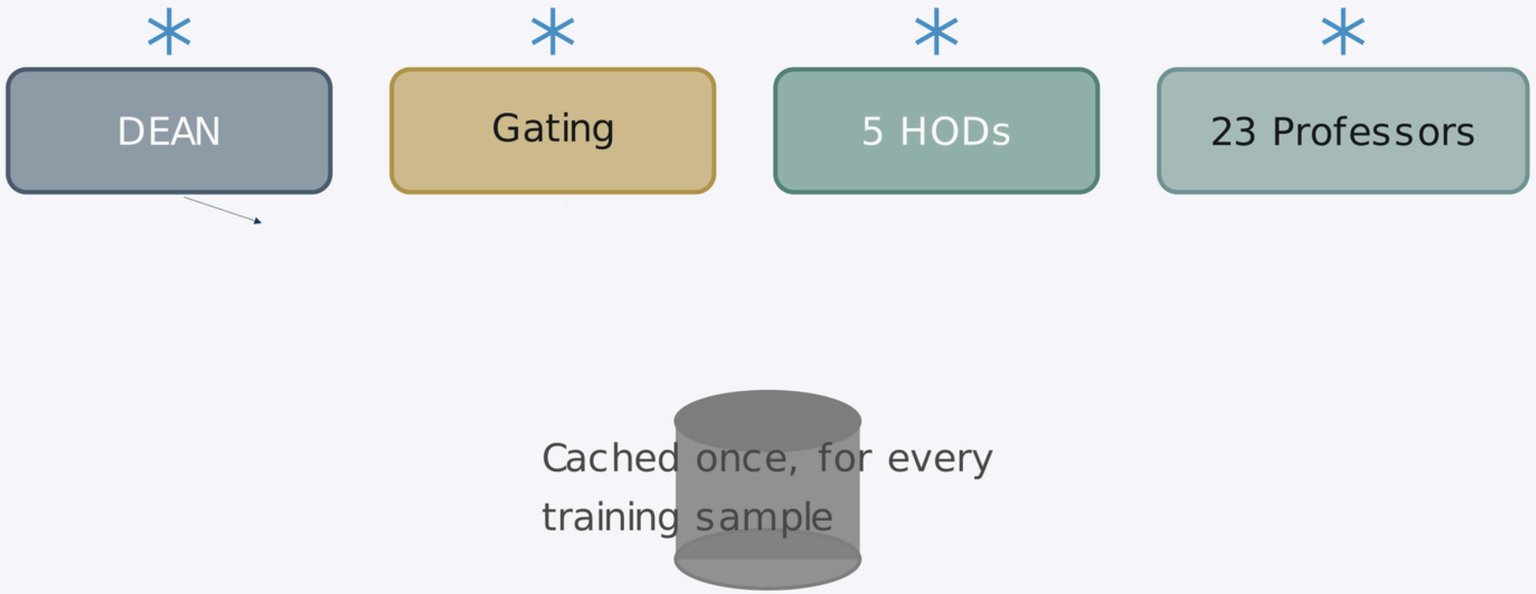
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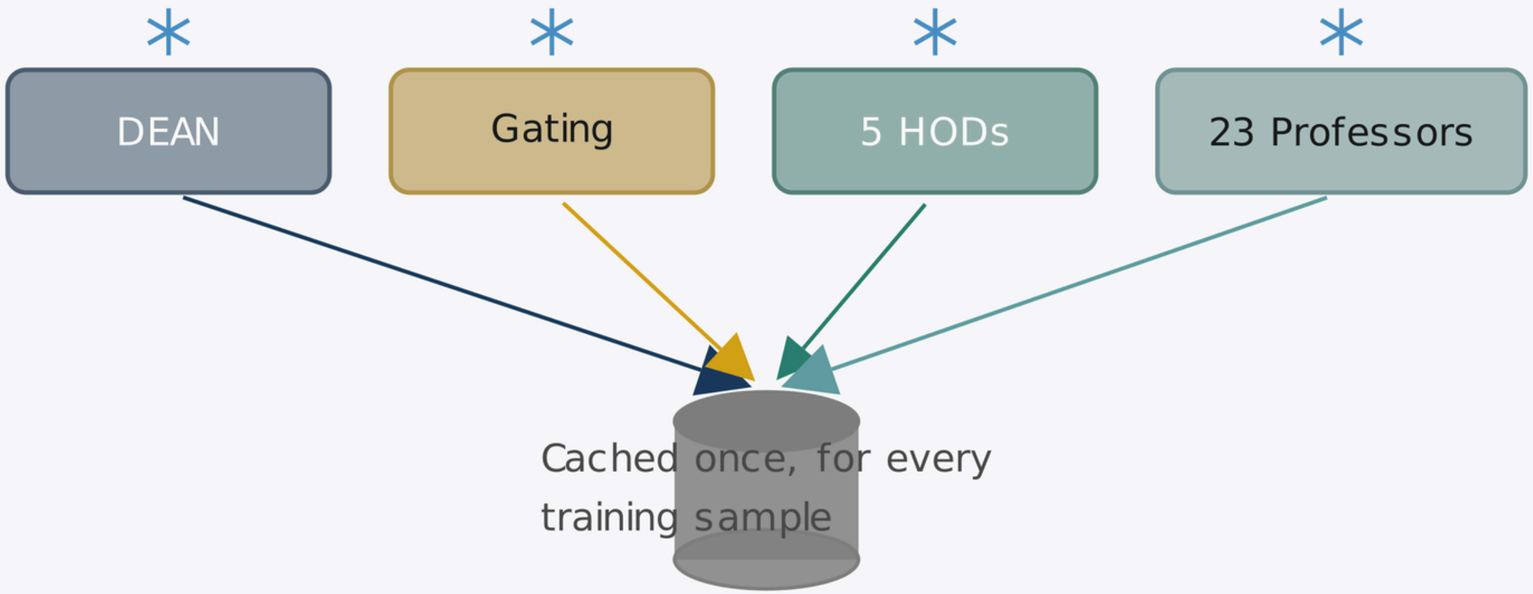
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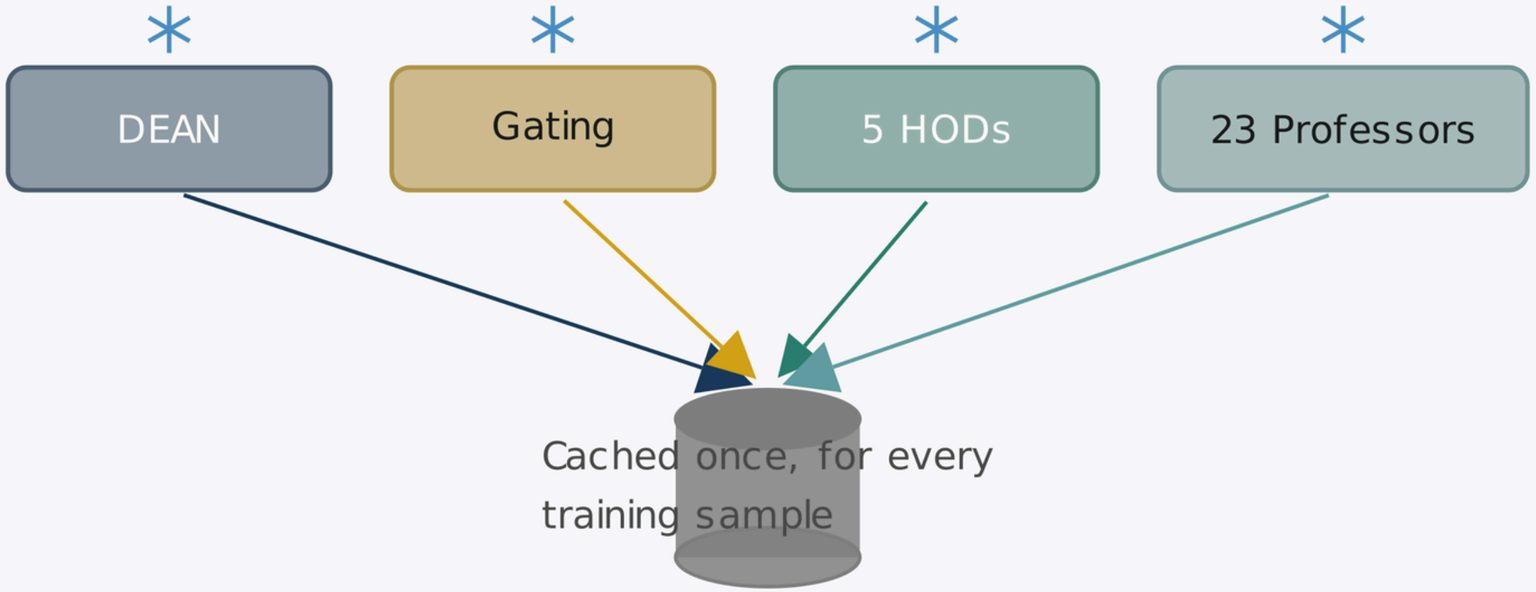
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Teacher-Output Caching



Stage 7

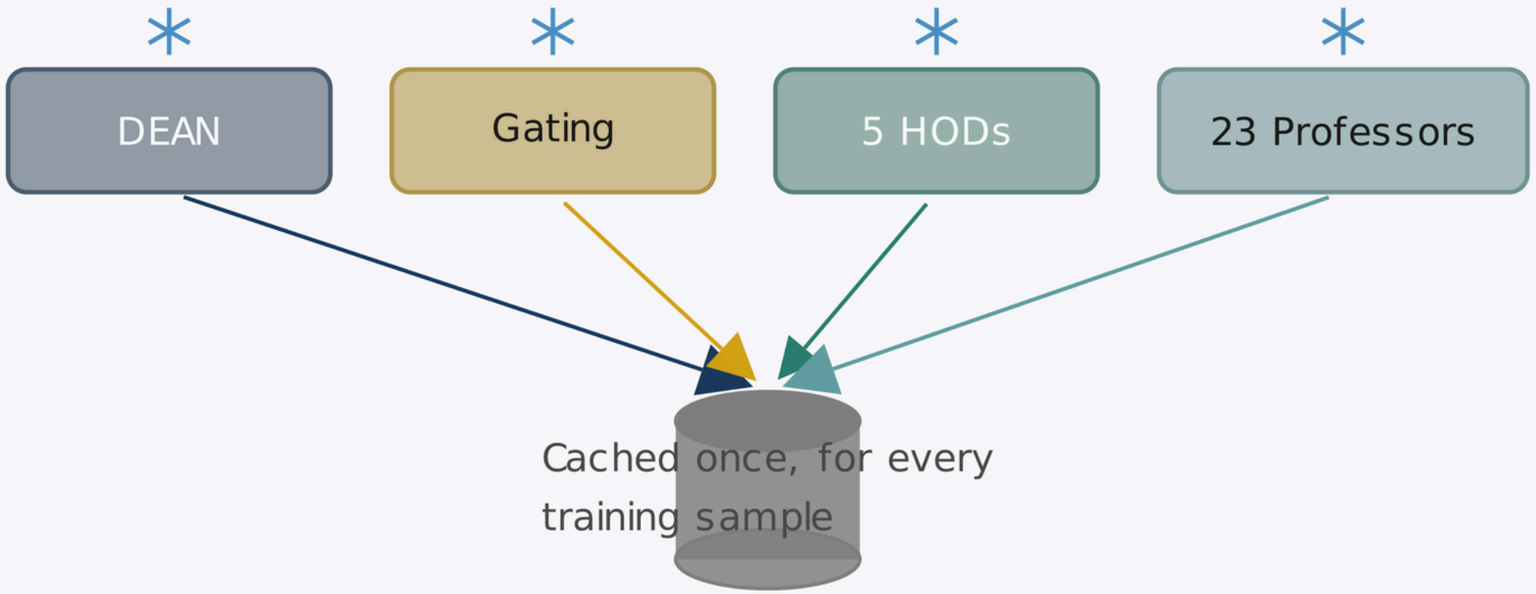
Teacher-Output Caching



- Professor per-subclass logits
- Professor penultimate embeddings
- Gating routing distributions g(x)
- DEAN's SSM output sequences

Stage 7

Teacher-Output Caching



- Professor per-subclass logits
- Professor penultimate embeddings
- Gating routing distributions g(x)
- DEAN's SSM output sequences

Stage 7: every teacher output is cached once, enabling efficient student training.

Stage 8: Chief Resident Skip-Distillation

Chief Resident - under 900K parameters

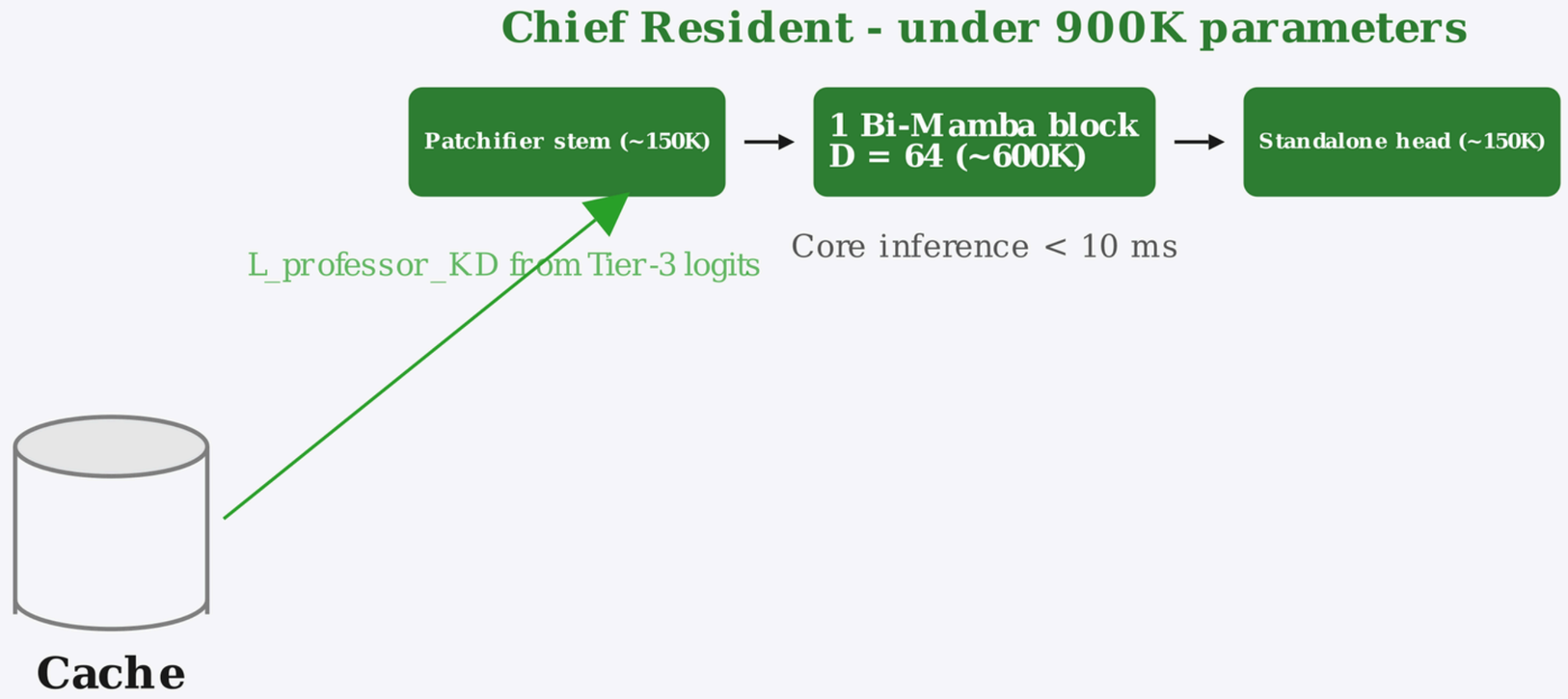


Core inference < 10 ms

Three skip-distillation terms plus a training-only consultation.

[11] Hinton et al., 2015 · [12] Park et al., 2019 · [13] Yang et al., 2023 · [15] Mirzadeh et al., 2020 · [16] Beigzadeh & Fathi, 2026

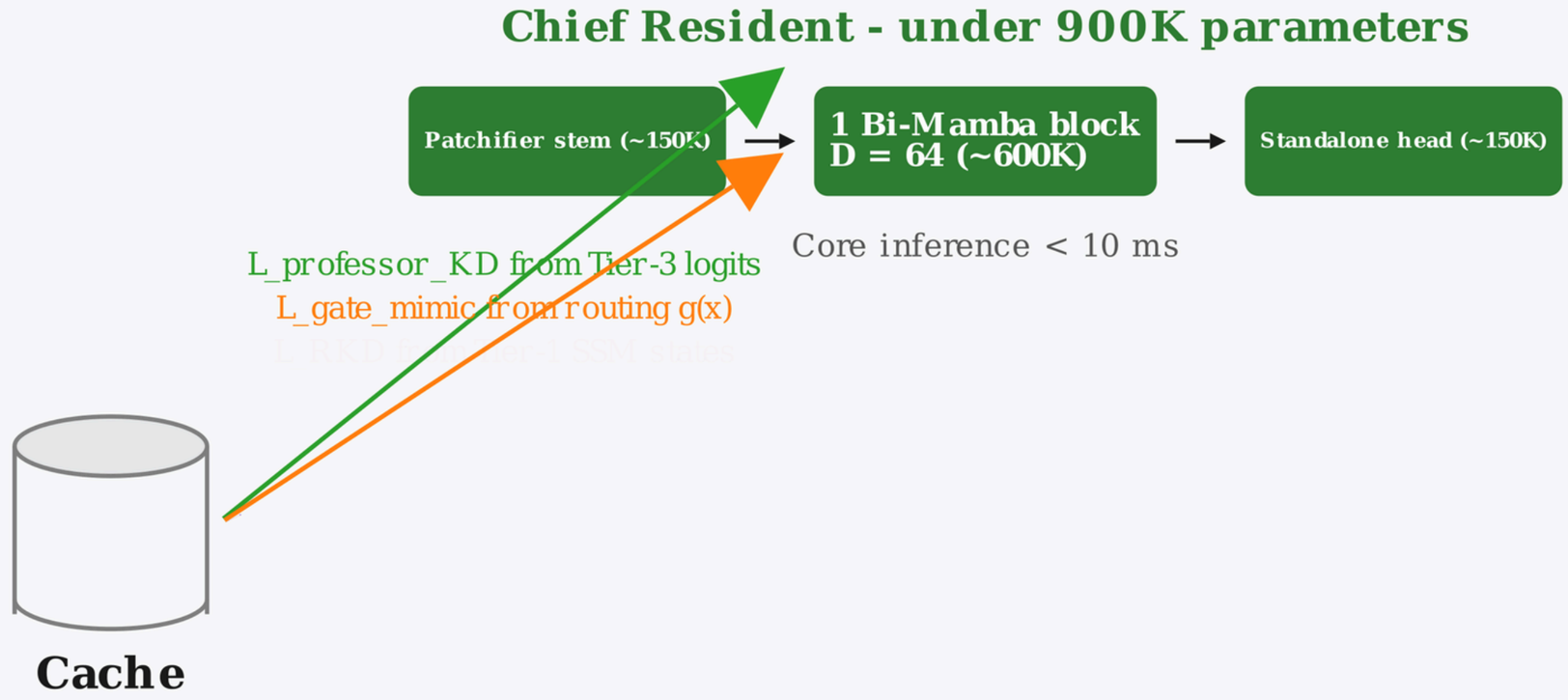
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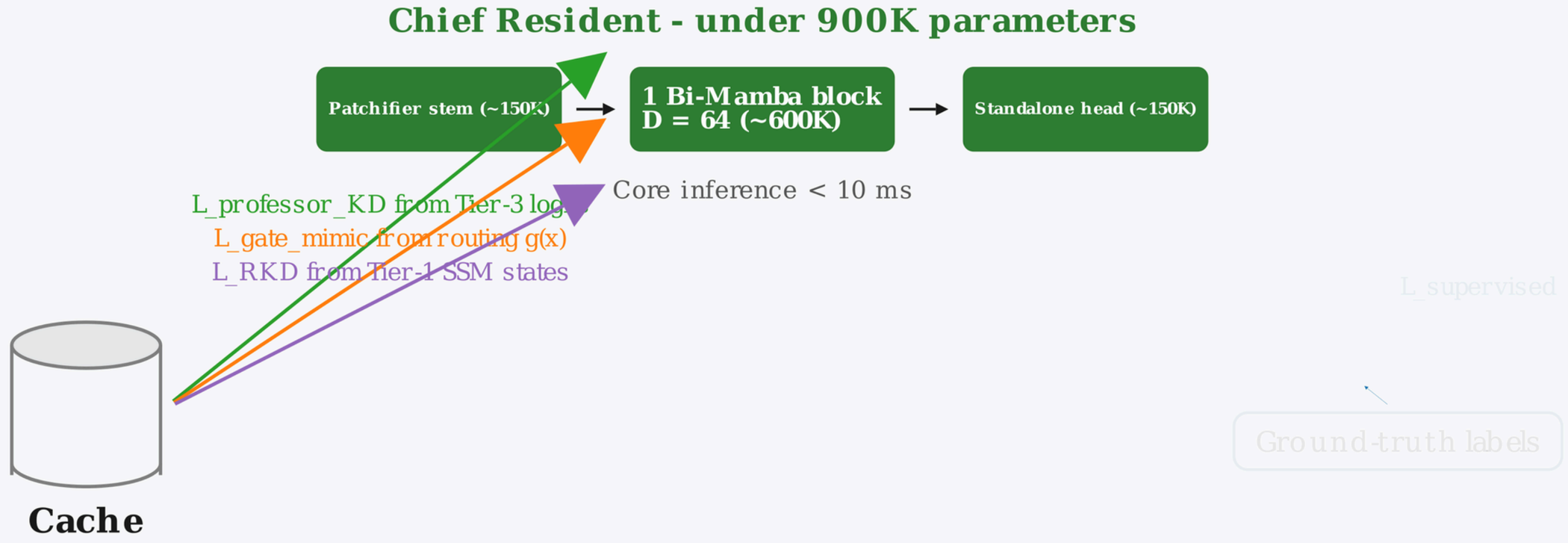
Stage 8: Chief Resident Skip-Distillation



Three skip-distillation terms plus a training-only consultation.

[11] Hinton et al., 2015 · [12] Park et al., 2019 · [13] Yang et al., 2023 · [15] Mirzadeh et al., 2020 · [16] Beigzadeh & Fathi, 2026

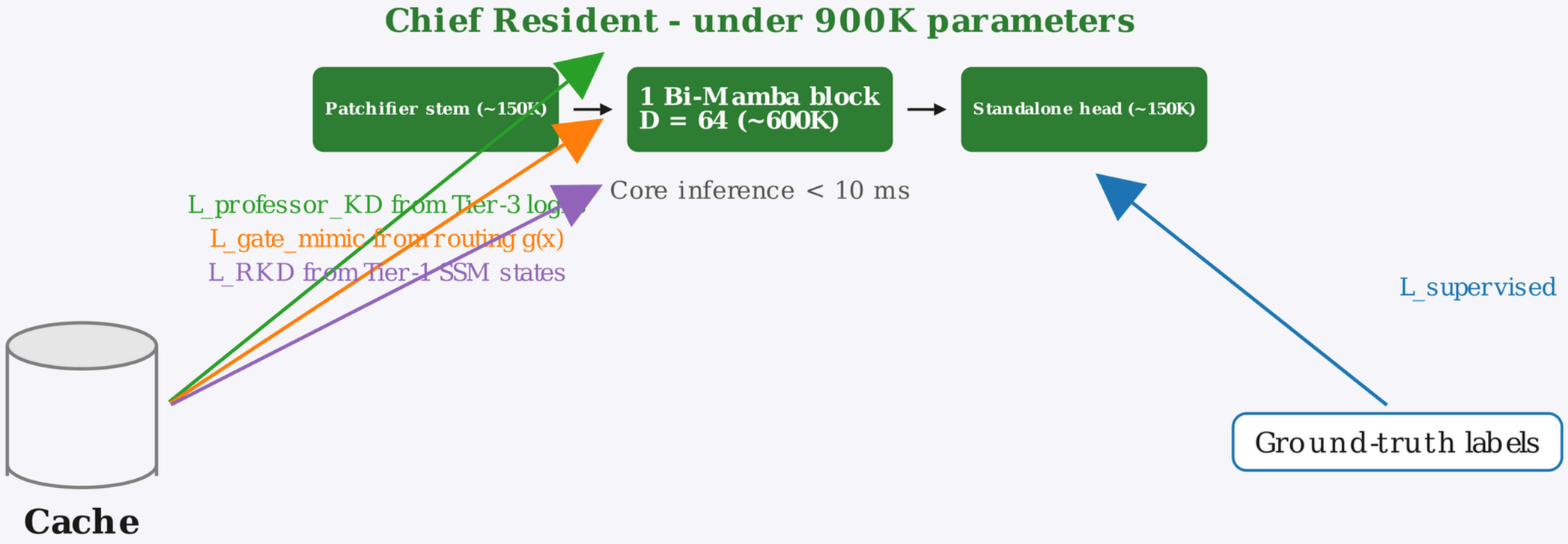
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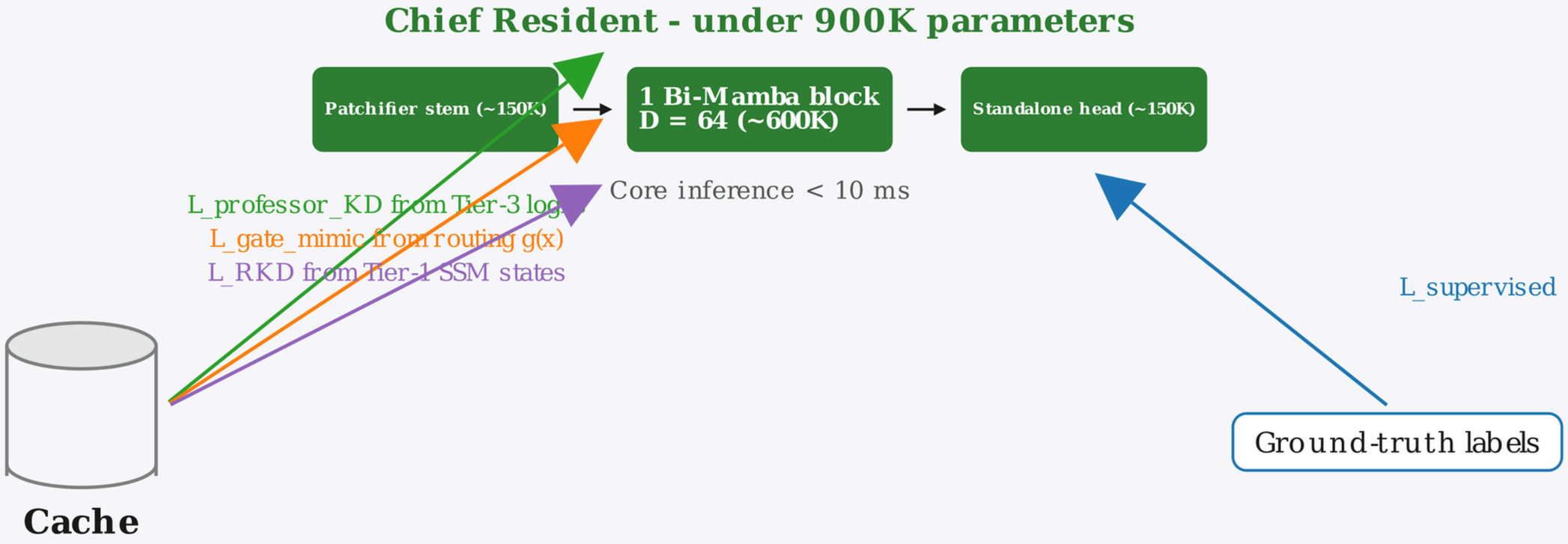
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Stage 8: Chief Resident Skip-Distillation



$$\mathcal{L}_{\mathrm{CR}} = \mathcal{L}_{\mathrm{sup}} + \lambda_1 \mathcal{L}_{\mathrm{profKD}} + \lambda_2 \mathcal{L}_{\mathrm{gate}} + \lambda_3 \mathcal{L}_{\mathrm{RKD}}$$

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Stage 8: Chief Resident Skip-Distillation

Chief Resident - under 900K parameters

Patchifier stem (~150K) → 1 Bi-Mamba block D = 64 (~600K) → Standalone head (~150K)

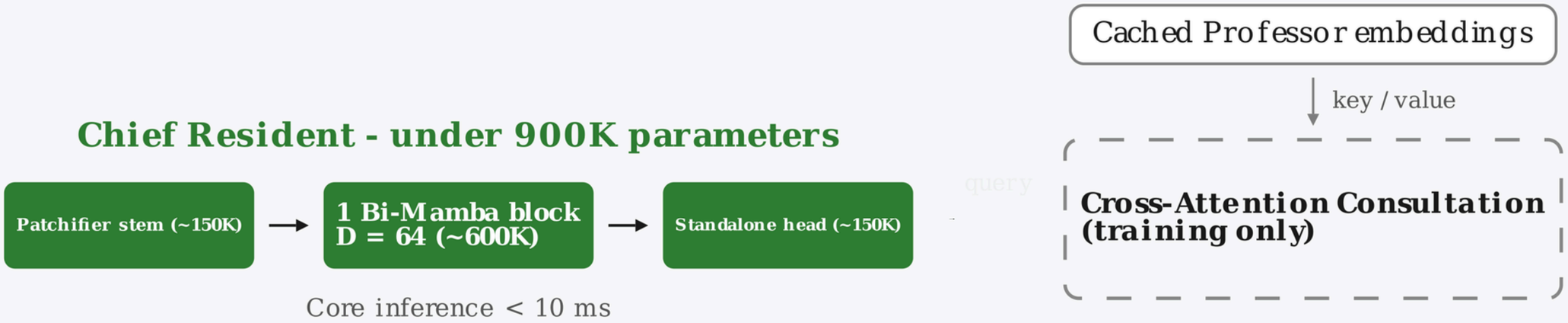
Core inference < 10 ms

Cross-Attention Consultation (training only)

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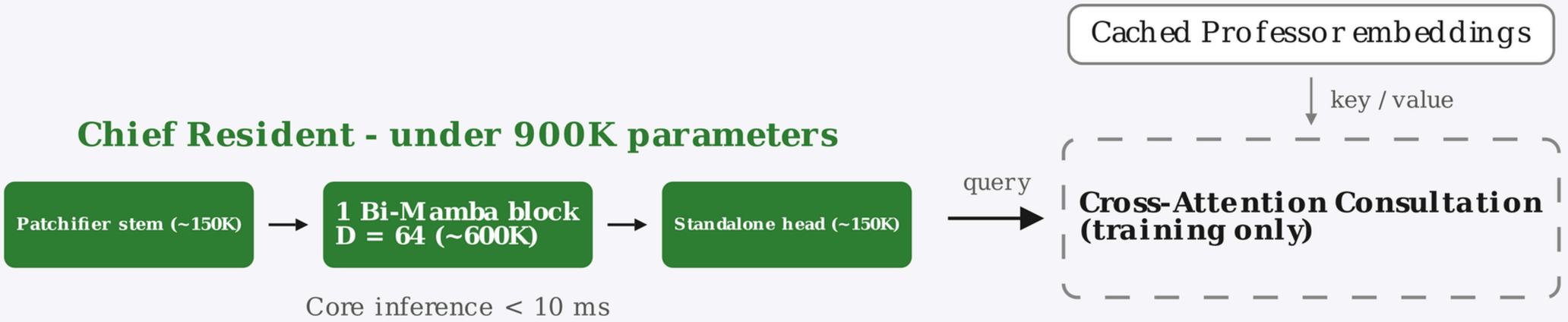
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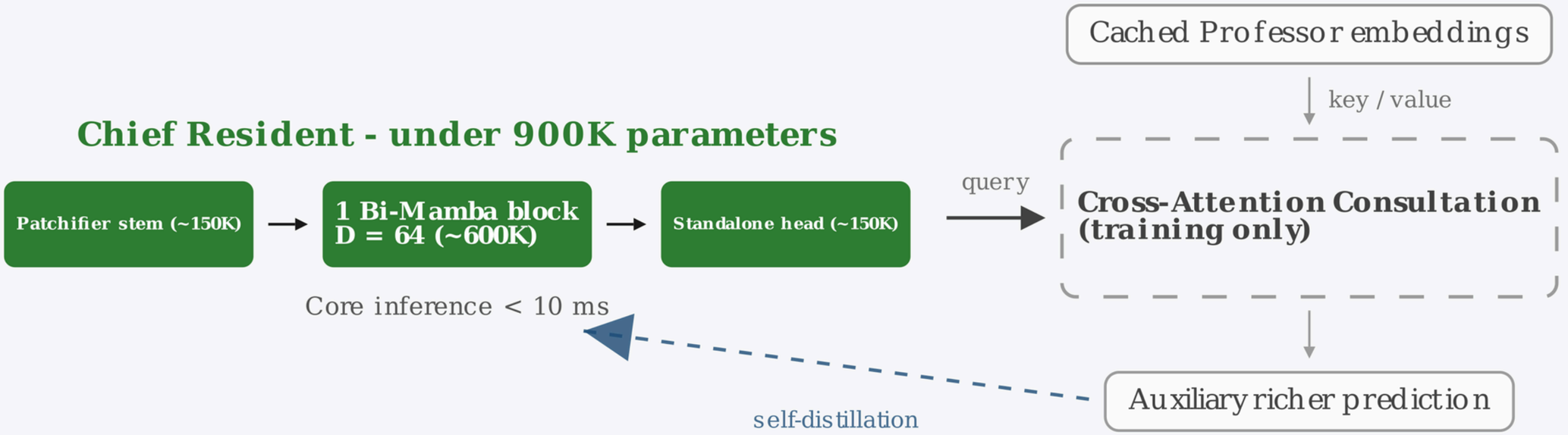
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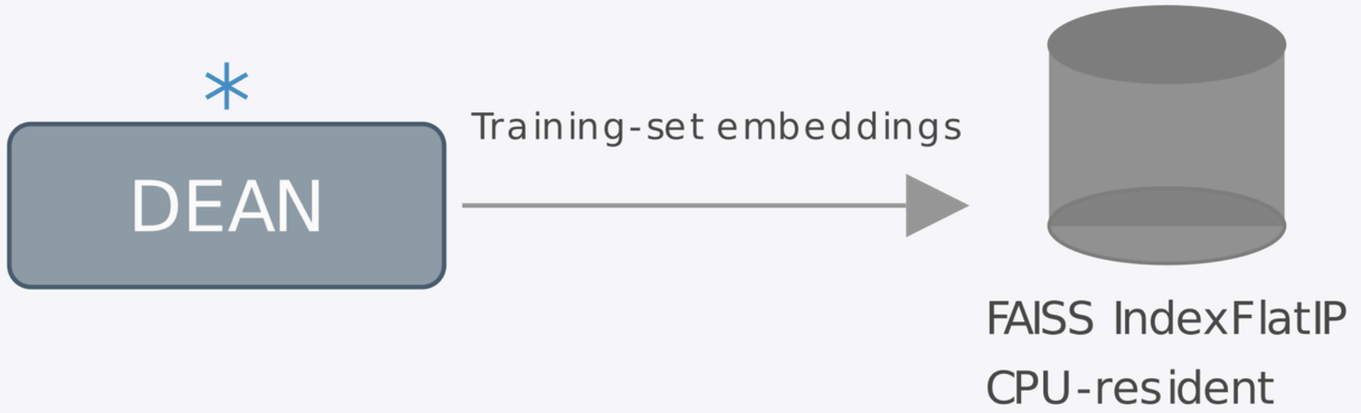
Stage 9

Prototype Index Construction

DEAN

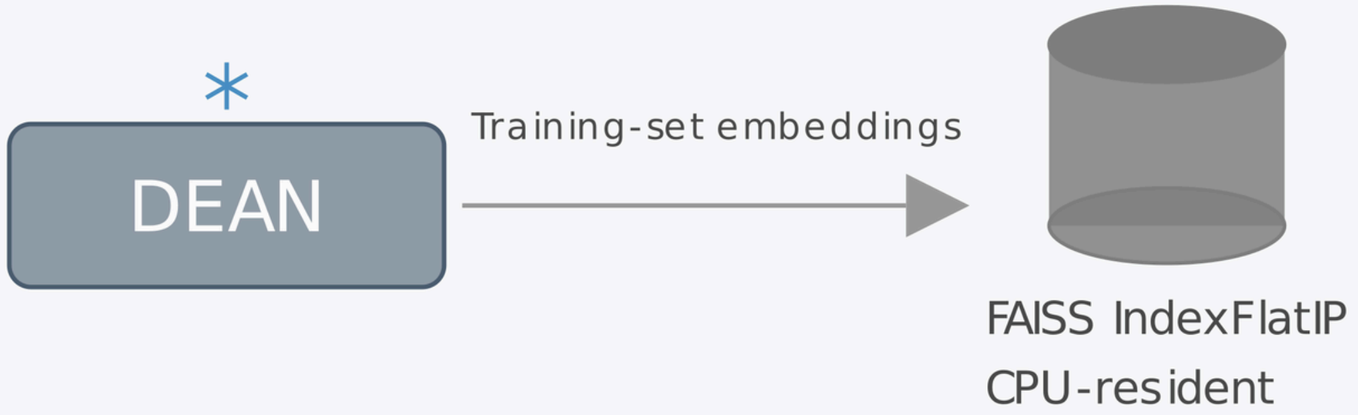
Stage 9

Prototype Index Construction



Stage 9

Prototype Index Construction

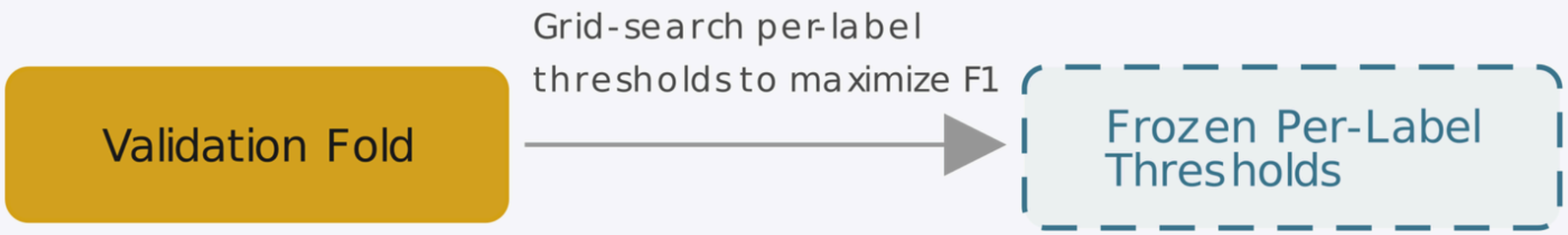


Built independently of Stages 4-8, branching off right after DEAN is frozen (Stage 3)

Stage 9: a FAISS index built from DEAN embeddings, for prototype retrieval.

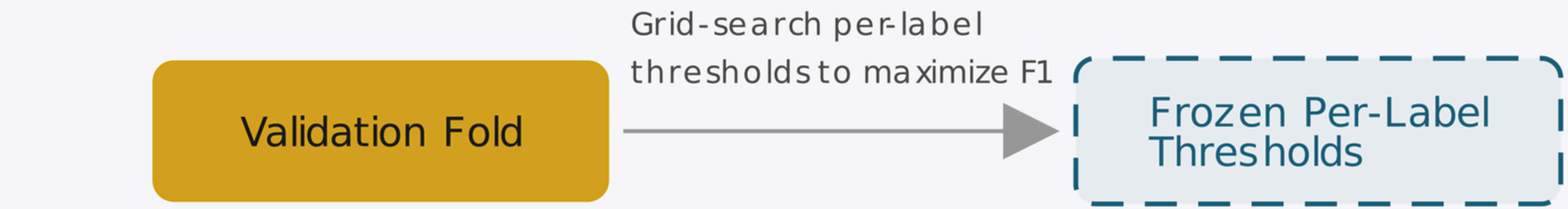
Stage 10

Calibration and Evaluation



Stage 10

Calibration and Evaluation



Test Fold
(used exactly once)

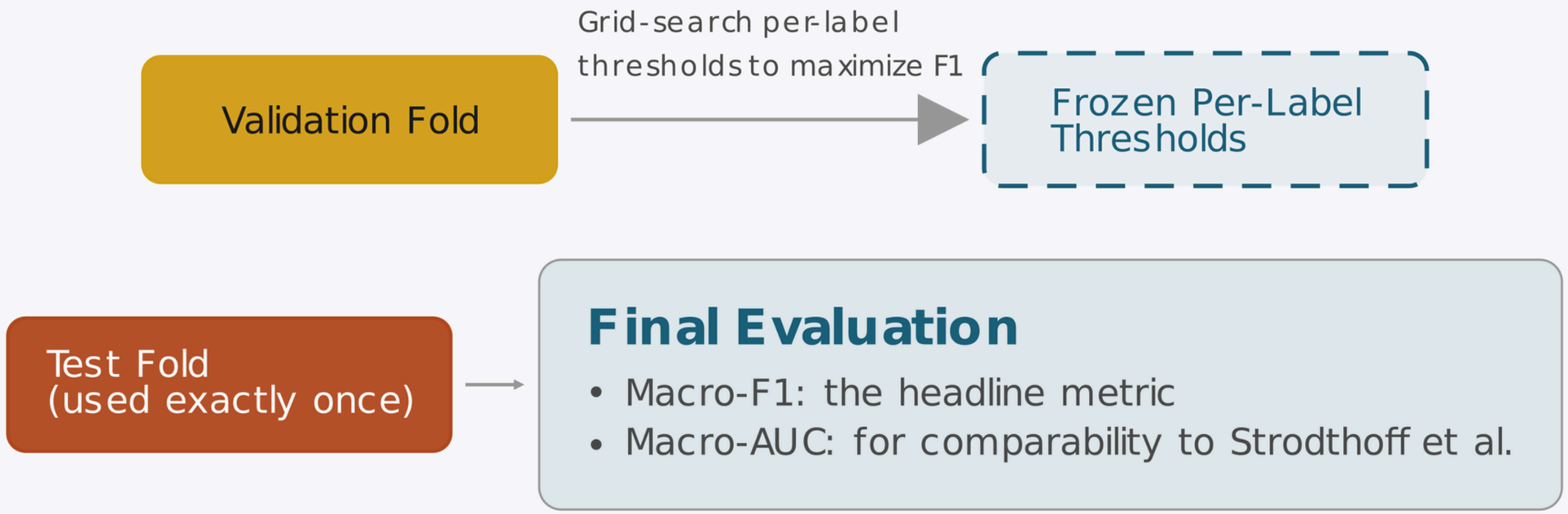
Final Evaluation

- Macro-F1: the headline metric
- Macro-AUC: for comparability to Strodthoff et al.

Core-model latency: under 10 ms (end-to-end latency, with FAISS lookup, is also benchmarked)

Stage 10

Calibration and Evaluation



Core-model latency: under 10 ms (end-to-end latency, with FAISS lookup, is also benchmarked)

Stage 10: thresholds calibrated on validation, evaluated once on held-out test.

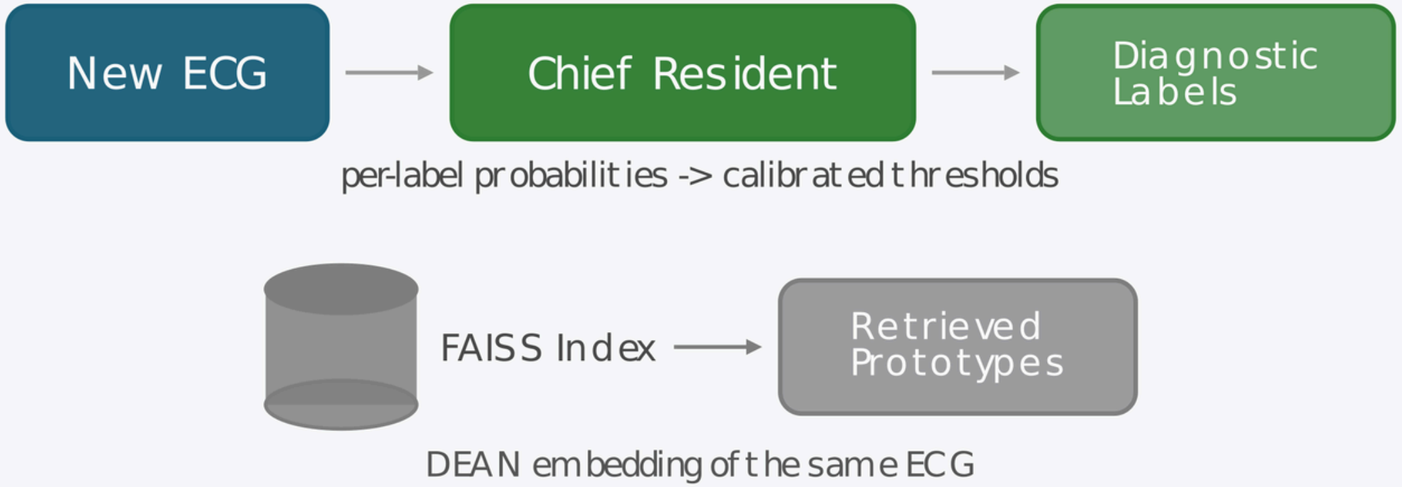
Deployment

Inference Flow



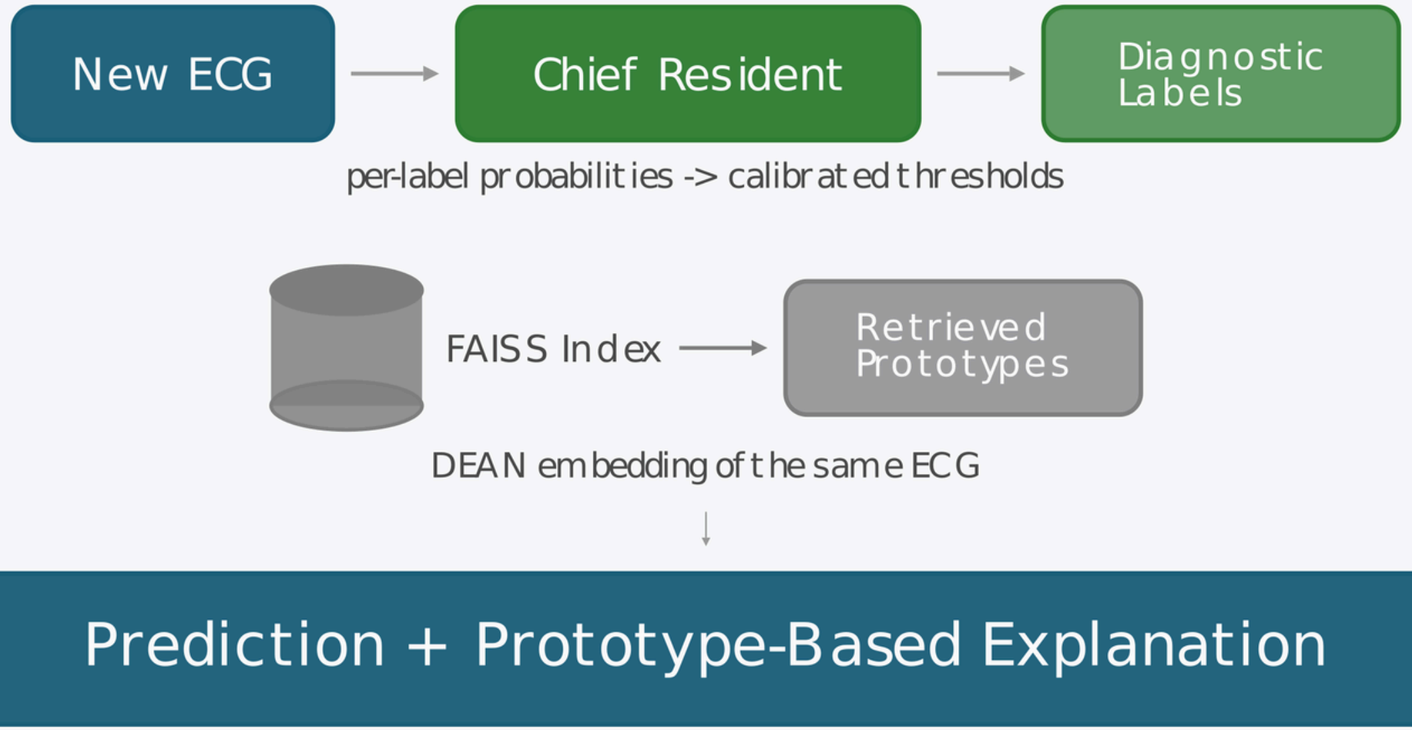
Deployment

Inference Flow



Deployment

Inference Flow



Only Chief Resident and FAISS run at inference

Deployment: only Chief Resident and FAISS are needed at runtime.

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